

DRC 常见 ERROR 与解决方案

1. 现阶段可 waive 的 DRC Error

可以 waive 的 DRC Error 列表如下

Check / Cell	Results	Flat
Check CO.R.4	1	9
Check ESD.24g	30	960
Check ESD.25g	1000	32000
Check LUP.3.3g_3.3V	1	32
Check LUP.5.3g_3.3V	3	96
Check MOM.R.2	1	1
Check DRM.R.1	1	1

另外 ESD.WARN.4 也可 waive

1. 确定可 waive 的 DRC Error

1.1 MOM.R.2

MOM.R.2 { @ Each MOM cell must be covered by MOMDMY_n (n=145;1~6/100). DRC only flags no MOMDMY_n (n=145;1~6/100) in the chip. But if there is no MOM cell in your chip, the violation can be waived.

MERGE (CHIP NOT INTERACT MOMDMY_ALL)

}

由于我们是纯数字电路设计，并没有使用到 MOM，因此可 waived。

1.2 ESD.24g

ESD.24g { @ The minimum width of RPO on drain side (X) for 5V,3.3V, 2.5V, 1.8V and 1.5V NMOS and PMOS.

@ exclude 1.8V and 1.5V NMOS when used as power clamp device $\geq 1.95\mu\text{m}$

$X = ((\text{ENSD_WCO OR EPSD_WCO}) \text{ AND ESD2DMY}) \text{ AND RPO}$

$Y = \text{EGTE COIN OUTSIDE EDGE X}$

$Z = \text{RPO COIN INSIDE EDGE X}$

$\text{ENC YZ} < 1.95 \text{ ABUT} < 90 \text{ REGION}$

}

IO 的 release note 中说明该 DRC ERROR 可 waive

来源: lib/IO/Documentation/release_note/rm_tpd018nv_280a.pdf

1.3 ESD.25g

ESD.25g { @ The minimum clearance from poly edge to CO edge on source side for NMOS and PMOS (Except ESD.26g) $\geq 0.5\mu\text{m}$

$X = \text{ECOS AND } ((\text{ENMOS OR EPMOS}) \text{ INTERACT RPO})$

$\text{EXT X EGTE_W} < 0.5 \text{ ABUT} < 90 \text{ OPPOSITE REGION}$

$Y = \text{ECO AND ESD1_NCS}$

$\text{EXT Y EGTE_W} < 0.5 \text{ ABUT} < 90 \text{ OPPOSITE REGION}$

}

IO 的 release note 中说明该 DRC ERROR 可 waive

来源: lib/IO/Documentation/release_note/rm_tpd018nv_280a.pdf

数字集成电路设计与实践

2. Seal Ring 相关的 DRC ERROR

2.1 SR.S.1

SR.S.1 { @ Minimum width of Assembly isolation >= ^SR_S_1 um

ASS_REGION AND COi

ASS_REGION AND DNWi

ASS_REGION AND NWELi

ASS_REGION AND POLYi

ASS_REGION AND DPO

ASS_REGION AND ODi

ASS_REGION AND DOD

ASS_REGION AND VIA1i

ASS_REGION AND VIA2i

ASS_REGION AND VIA3i

ASS_REGION AND VIA4i

ASS_REGION AND VIA5i

ASS_REGION AND M1i

ASS_REGION AND M2i

ASS_REGION AND M3i

ASS_REGION AND M4i

ASS_REGION AND M5i

ASS_REGION AND M6i

ASS_REGION AND VIADi

ASS_REGION AND MDi

ASS_REGION AND UBMi

ASS_REGION AND CBDi

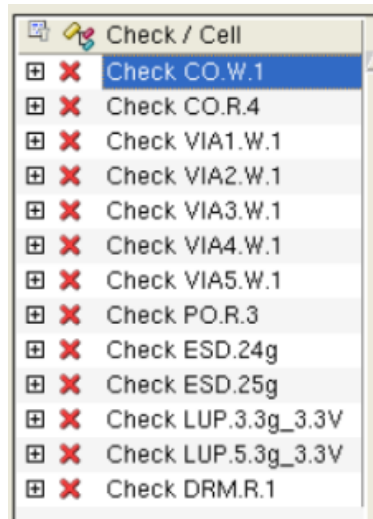
ASS_REGION AND CBi

}

DRC 要求芯片与 Seal Ring 之间最小要有 5um 的间隔，当间隔不足 5um 时则会出现该 Error，需要重新修改 Seal Ring，增大间隔。

2.2 DRC rule file 中 Seal Ring 选项未打开

出现下面的 ERROR 时，一般是 DRC rule file 中的选项不对，需要将其中的 FULL_CHIP 和 WITH_SEALRING 的注释删掉



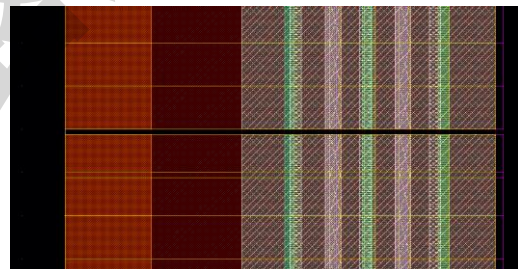
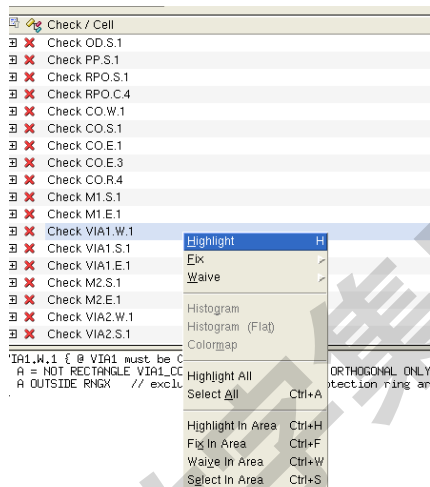
```

'61 VARIABLE VSS_TEXT "VSS" "VSSPST" // pin name of VSS
'62
'63 #DEFINE FULL_CHIP // Turn on for chip level design of sealring and dummy PAD, otherwise please turn off it.
'64
'65 // #DEFINE DENSITY_IN_BLOCKAGE // turn on only when to check density within blockage layers
'66 // #DEFINE BigSealringCorner // Turn on to use big seal-ring corner for whole chip design
'67 // #DEFINE SmallSealringCorner // Turn on to use small seal-ring corner for whole chip design
'68 #DEFINE WITH_SEALRING // Turn on to check sealring rules
'69
'70 // #DEFINE ALL_AREA_IO // Turn on for treat OD injector as same as AAIO
'71
'72

```

2.2 Seal Ring 重叠或缝隙

Seal Ring 不允许出现重叠或缝隙，若出现则会出现大量错误，类似下图。可以右键具体 ERROR，然后 highlight，即可找到出现问题的地方



2.3 PO.R.5 或 PO.R.3

参考 DRC Error 解决-PO.R.5-更新 1009.pdf

2.4 未加 Dummy OD

Check / Cell	Results	Flat
Check CO.R.4	1	9
Check PO.R.3	1	1
Check ESD.24g	30	960
Check ESD.25g	1000	32000
Check LUP.3.3g_3.3V	1	32
Check LUP.5.3g_3.3V	3	96
Check DOD.R.1	1	1
Check PO.R.5	1	1
Check MOM.R.2	1	1
Check DRM.R.1	1	1

未加 Dummy OD 时，会出现多个 Poly 和 DIFF 的密度问题。补上 Dummy OD 即可解决该问题。

2.5 未加 Dummy Metal

未加 Dummy metal 时，会出现多个 Metal 的 density，类似 M2.R.1，M3.R.1，等。补上 Dummy Metal 即可解决该问题。

2.6 DOD.E.2 和 DPO.E.1

Filter: Show Unresolved

final_gds2, 2716 Results (in 7 of 444 Checks)

Check / Cell	Results	Flat
Check CO.R.4	1	9
Check PO.R.3	1	1
Check ESD.24g	30	960
Check ESD.25g	1000	32000
Check LUP.3.3g_3.3V	1	32
Check LUP.5.3g_3.3V	3	96
Check DOD.E.2	683	
Check DPO.E.1	1000	
Check MOM.R.2	1	1
Check DRM.R.1	1	1

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